

Multiple Access Techniques

Visual comparison of telecommunication channel access methods



FDMA

Frequency Division Multiple Access



TDMA

Time Division Multiple Access



CDMA

Code Division Multiple Access



OFDMA

Orthogonal Frequency Division Multiple Access

▶ Play Animation

FDMA

Users are separated by frequency bands



Advantages

- ✓ Simple implementation
- ✓ No synchronization needed
- ✓ Constant bandwidth per user

Disadvantages

- ✗ Inflexible bandwidth allocation
- ✗ Guard bands waste spectrum
- ✗ Cannot handle variable traffic

Key Insight

FDMA divides the spectrum into frequency bands, like different radio stations. Each user gets their own frequency channel.

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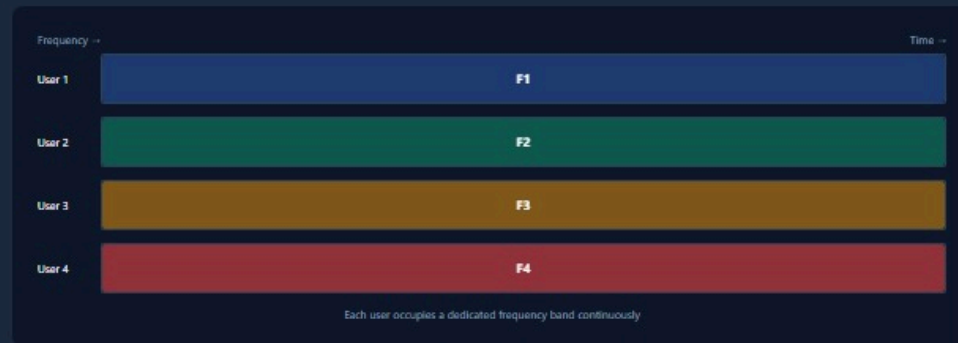
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■ User 1 ■ User 2 ■ User 3 ■ User 4

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TDMA

Users share frequency but get different time slots

Same Frequency, Different Time Slots



Users transmit in sequential time slots on the same frequency

■ User 1 ■ User 2 ■ User 3 ■ User 4

Advantages

- ✓ Flexible time slot allocation
- ✓ No frequency guard bands
- ✓ Better spectral efficiency than FDMA

Disadvantages

- ✗ Requires precise synchronization
- ✗ Higher peak power needed
- ✗ Delay sensitive

Key Insight

TDMA divides time into slots. Users take turns transmitting in their assigned time slots, like a round-robin conversation.

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CDMA

Users share time and frequency with unique codes

All users transmit simultaneously with unique codes

Overlapping Signals

Separated by unique spreading codes

Code 1

01020011

Code 2

10001110

Code 3

11101111

Code 4

11010011

Each user has a unique orthogonal code for signal separation

■ User 1 ■ User 2 ■ User 3 ■ User 4

Advantages

- ✓ High capacity
- ✓ Soft handoff capability
- ✓ Better privacy and security

Disadvantages

- ✗ Near-far problem
- ✗ Complex receivers
- ✗ Self-interference issues

① Key Insight

CDMA allows all users to transmit simultaneously on the same frequency using unique spreading codes, like multiple conversations in different languages.

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Users share time and frequency with unique codes

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Overlapping Signals

Separated by unique spreading codes

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10111100

Code 2

00101010

Code 3

10101010

Code 4

01110111

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OFDMA

Dynamic allocation of frequency-time resource blocks

— Frequency Subcarriers —

Time Slot 1



Resource blocks (time-frequency grid) dynamically allocated to users

■ User 1 ■ User 2 ■ User 3 ■ User 4

Advantages

- ✓ Excellent spectral efficiency
- ✓ Robust against interference
- ✓ Flexible resource allocation

Disadvantages

- ✗ High PAPR
- ✗ Sensitive to frequency offset
- ✗ Complex implementation

Key Insight

OFDMA combines benefits of FDMA and TDMA by creating a grid of time-frequency resource blocks that can be dynamically allocated, like a smart scheduling system.

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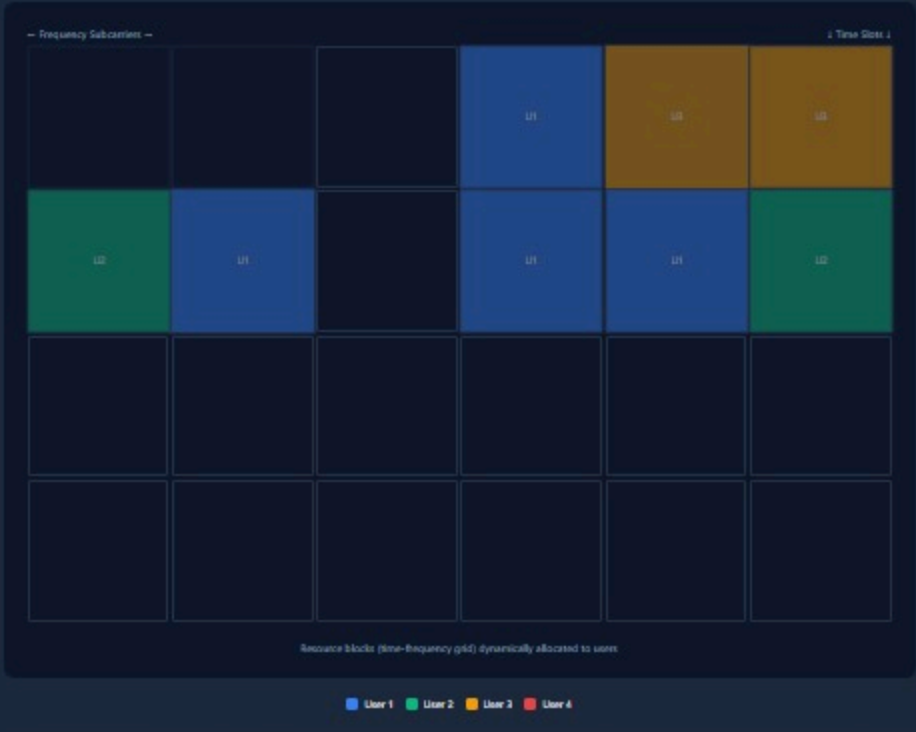
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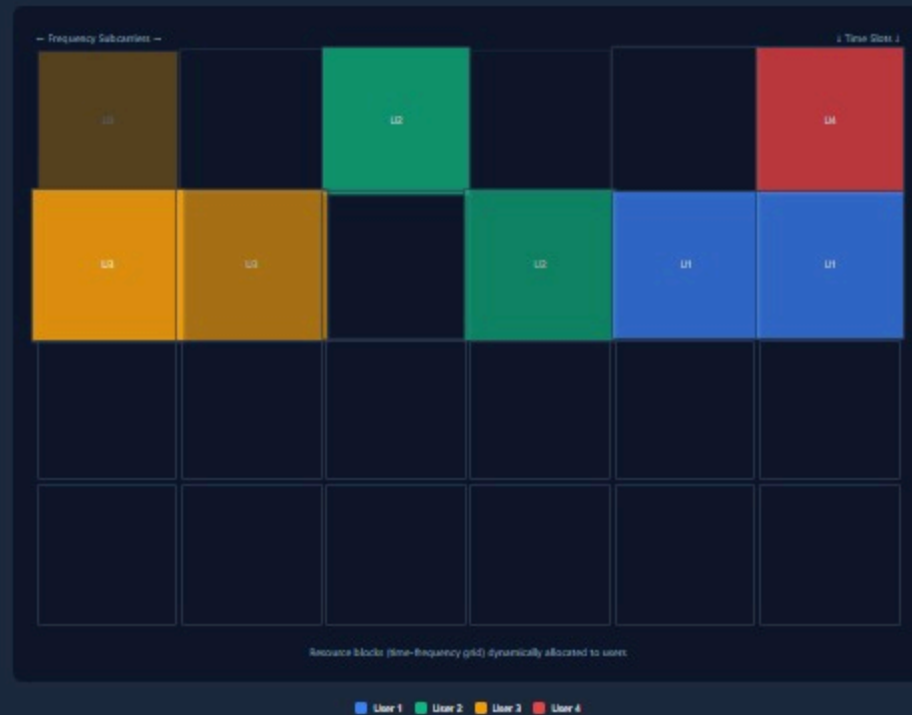
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